Math 237A HW 6

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1 D

Problem 1

> Lazarsfeld Problem Set 5 (2):
> Let $f : X \to Y$ be a dominant morphism of irreducible (quasi-projective) varieties. Prove that $\dim Y \leq \dim X$.

Solution: We'll first work with the affine case: Given that $f : X \to Y$ is a dominant morphism, then the induced ring homomorphism $f^* : k[Y] \to k[X]$ is injective. Which, such ring homomorphism can be extended to the field homomorphism between their fraction fields, say $\overline{f^*} : k(Y) \to k(X)$ (or between their field of rational functions).

Now, suppose $\dim Y = n$, then there exists an algebraically independent subset over k with length n, say $g_1, ..., g_n \in k(Y)$. So, the field homomorphism $\overline{f^*}$ must have $\overline{f^*}(g_1), ..., \overline{f^*}(g_n) \in k(X)$ be algebraically independent also:

Suppose the contrary that they're algebraically dependent, then there exists $p(x_1, ..., x_n) \in k[x_1, ..., x_n]$, such that $p\left(\overline{f^*}(g_1), ..., \overline{f^*}(g_n)\right) = 0$. Then, since $\overline{f^*}$ is a field homomorphism, one has $0 = p\left(\overline{f^*}(g_1), ..., \overline{f^*}(g_n)\right) = \overline{f^*}(p(g_1, ..., g_n))$, showing $p(g_1, ..., g_n) = 0$ by the injectivity of field homomorphism. So, $g_1, ..., g_n$ is no longer algebraically independent over k, which is a contradiction.

So, $\overline{f^*}(g_1), ..., \overline{f^*}(g_n) \in k(X)$ is algebraically independent over k, showing $n \leq \operatorname{trdeg}_k k(x) =: \dim X$. Hence, $\dim Y \leq \dim X$.

Finally, for the quasi-projective case, it suffices to cover using affine charts, and provide the statement above the complete the claim for each affine charts, which provides a general statement for the whole quasi-projective variety also.

2 D

Problem 2

Lazarsfeld Problem Set 5 (3):

Given $d \geq 2$, denote by $\mathbb{P}^{N(d)}$ the projective space parametrizing all plane curves of degree d. (So $N(d) = \binom{d+2}{2} - 1$).

(a) Show that if $d = 2$, then there is a dense open subset $U \subset \mathbb{P}^5$ such that all conics corresponding to points in U are projectively equivalent, i.e. differ by a linear change of coordinates.

(b) On the other hand, prove that the analogous statement fails when $d \geq 3$.

Note: You may grant that $\mathrm{SL}(n+1)$ is irreducible. You will want to observe that it has dimensoin $= (n+1)^2 - 1$.

Solution:

(a) Recall that given $\mathbb{P}^5 = \{[a, b, c, d, e, f]\}$ can also be identified as all the matrix $S = \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}$ quotient out by scalar multiplication of $k^\times$ (which each represents the conic $ax^2 + by^2 + cz^2 + dxy + exz + fyz$ over $\mathbb{P}^2$). Then, if using $\mathrm{SL}(3)$ as a source of projective transformation on $\mathbb{P}^2$ (which transforms the coordinates, hence the coefficients of the conic over $\mathbb{P}^2$), then for any $A \in \mathrm{SL}(3)$, the transformation on the coefficients of the conics can be provided by $A^T S A$ as matrices.

Now, if consider the morphism $\varphi : \mathrm{SL}(3) \to \mathbb{P}^5$ by $A \mapsto A^T I A = A^T A$ (or, doing a coordinate transformation on a standard conic $x^2 + y^2 + z^2$), then we have $\operatorname{im}(\varphi) \subseteq \left\{ \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix} \middle| \det \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix} \neq 0 \right\}$ if identify $\mathbb{P}^5$ as symmetric matrices quotient out by scalar multiplication (since $\det(A^T A) \neq 0$ for all $A \in \mathrm{SL}(3)$). Now, notice that $\operatorname{im}(\varphi) = \left\{ \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix} \middle| \det \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix} \neq 0 \right\}$, since for any such symmetric matrix S (with $\det S \neq 0$), it is diagonalizable, so there exists $P \in \mathrm{GL}(3)$ (with $P^{-1} = P^T$) and diagonal matrix $D \in \mathrm{GL}(3)$, such that $S = P^T D P$. Then, take the square root matrix $\sqrt{D}$ (which is also diagonal; this is defined due to the fact that $x^2 = \lambda$ always has a solution over $k = \overline{k}$), one has $S = P^T D P = P^T \sqrt{D}^T \sqrt{D} P = \left(\sqrt{D} P\right)^T \left(\sqrt{D} P\right)$ (where $\sqrt{D} P \in \mathrm{GL}(3)$), then since S is up to a scalar, by suitable scaling factor one can have $\sqrt{D} P \in \mathrm{SL}(3)$, showing that the morphism $\varphi : \mathrm{SL}(3) \to \mathbb{P}^5$ is in fact dominant (since the image is an open dense subset, where the symmetric matrix has determinant nonzero).

Since within this open dense subset of $\mathbb{P}^5$, everything is a conic after a linear transformation on the coordinate of $x^2 + y^2 + z^2$, we can claim that within this open dense image, everything iis projectively equivalent (since it's up to a change of coordinates using elements of $\mathrm{SL}(3)$).

(b) Given $d \geq 3$, notice that $\binom{d+2}{2} = \frac{(d+2)(d+1)}{2} \geq \frac{5 \cdot 4}{2} = 10$, $\mathbb{P}^{N(d)}$ (the space of all plane curves of degree d) has dimension ≥ 10. But, if using the same method as in part (a), since $\dim \mathrm{SL}(3) = 8$, there is no dense subset in $\mathbb{P}^{N(d)}$ so that $\mathrm{SL}(3)$ can map surjectively onto through a morphism like in part (a), simply because the dimension of the image must have dimension ≤ 8, which has closure at most dimension 9 (hence the closure cannot cover the whole space, since the space has dimension ≥ 10). Therefore, the image cannot be dense, showing such claim fails.

3 D

Problem 3

Lazarsfeld Problem Set 5 (4):
Find the dimension of the space $M_{n\times m}^{\leq r}$ of all $n\times m$ matrices of rank $\leq r$.

Solution: Given any matrix $A\in M_{n\times m}^{=r}$ (an open dense subset of $M_{n\times m}^{\leq r}$), since its column rank is r, if identify the $W_A\subseteq k^n$ as the column span of the columns of A, it is an r-dimensional subspace of k^n. Which, A can be identified as W_A, together with m column vectors $c_1,...,c_m\in W_A$, so it can also be thought of as a pair in $\mathbb{G}(r,n)\times(\mathbb{A}^{nm})$ (where $\mathbb{A}^{nm}:=M_{n\times m}$, all the $n\times m$ matrix).

More specifically, take $Z\subseteq\mathbb{G}(r,n)\times\mathbb{A}^{nm}$ with all the pairs (W,A) such that the column vectors of $A\in M_{n\times m}$, say $c_1,...,c_m\in W$ (where W is an r-dimensional subspace). This forms an algebraic set, since each r-dimensional subspace $W\subseteq\mathbb{A}^n$ is also an algebraic set cut out by several linear equations, and the requirement for (W,A) to form a pair is that all the column vectors of A $(c_1,...,c_m\in\mathbb{A}^n)$ must satisfy the associated equations of W.

Notice that fixing any $W\in\mathbb{G}(r,n)$, the subset $\mathrm{pr}_1^{-1}(W)=\{W\}\times W^m$ (where now view $\mathbb{A}^{nm}:=\bigoplus_{i=1}^m k^n$, and each $W\in k^n$), since for each pair (W,A), it can also be viewed as $(W,\{c_1,...,c_m\})$ (where each $c_i\in W$). THen, locally Z is a fibre bundle with fibre being $W^m\cong\mathbb{A}^{rm}$ (wince W is an r-dimensional subspace), showing $\dim Z=\dim\mathbb{G}(r,n)+\dim W^m=r(n-r)+mr=r(n+m-r)$.

Now, if consider the projection $\mathrm{pr}_2:Z\to\mathbb{A}^{nm}=M_{n\times m}$. Since every pair $(W,A)\in Z$ has $A\in M_{n\times m}$ being a matrix with column vectors in W (an r-dimensional subspace), then rank of A is $\leq r$ (since the column span is contained in W), showing $\mathrm{im}(\mathrm{pr}_2)\subseteq M_{n\times m}^{\leq r}$; and, the two are in fact equal as sets, since for any $A\in M_{n\times m}^{\leq r}$, take $W\subseteq k^n$ as an r-dimensional subspace contain the column span of A, then $(W,A)\in Z$, and $\mathrm{pr}_2(W,A)=A$. So, $\mathrm{pr}_2:Z\twoheadrightarrow M_{n\times m}^{\leq r}$ is a surjective morphism, showing that $\dim M_{n\times m}^{\leq r}\leq\dim Z=r(n+m-r)$.

Finally, to show they're in fact equal, we'll show it's a birational equivalence also: take $\mathrm{pr}_2^{-1}(M_{n\times m}^{=r})\subseteq Z$ a dense open subset (by continuity of pr_2), notice that the restriction onto this open subset $\mathrm{pr}_2:\mathrm{pr}_2^{-1}(M_{n\times m}^{=r})\to M_{n\times m}^{=r}$ is in fact a one-to-one correspondance: such map is surjective, based on the fact that when restricting the codomain to $M_{n\times m}^{\leq r}$, the second projection pr_2 is surjective. It's injective, since if $(W_A,A),(W_B,B)\in\mathrm{pr}_2^{-1}(M_{n\times m}^{=r})$ satisfies $A=B$ (or, their second projection are the same), then since A,B both have the same column span being r-dimensional, while W_A,W_B are also r-dimensional subspaces containing their column span, which enforces $W_A=W_B=$ column span of A,B (since the dimension matches).

Now, if restrict to an even smaller open subset, say $U\subseteq M_{n\times m}^{=r}$, such that all matrices have the first $r\times r$ minor with determinant $\neq 0$, then it implies for any $A\in U$, the first r column vectors form a basis for the column span, hence using Plucker coordinates on the first r columns (or on the left most $n\times r$ submatrix), it defines a rational map $\varphi:M_{n\times m}^{=r}\dashrightarrow\mathrm{pr}_2^{-1}(M_{n\times m}^{=r})$, by $\varphi(A)=(W_A,A)$, where W_A is the subspace generated by the first r columns (using Plucker coordinates, it's taking all $r\times r$ minor's determinant in the left most $n\times r$ submatrix of A, which is a rational map into $\mathbb{G}(r,n)$). And, this rational map is well-defined on $U\subseteq M_{n\times m}^{=r}$, since then the choice W_A is in fact an r-dimensional subspace (or a well-defined point in $\mathbb{G}(r,n)$ when taking the Plucker coordinates).

Then, we claim that $\varphi:U\to\mathrm{pr}_2^{-1}(U)$ defines an inverse of $\mathrm{pr}_2:\mathrm{pr}_2^{-1}(U)\to U$: Given any $(W,A)\in\mathrm{pr}_2^{-1}(U)$, one has W being the span of the first r column vectors of A, so $\mathrm{pr}_2(W,A)=A$, while $\varphi(A)=(W,A)$; Now, given any $B\in U$, it's clear that $\mathrm{pr}_2(\varphi(B))=B$ by definitions. So, they're inverse of each other as rational maps.

Hence, $\mathrm{pr}_2:Z\to M_{n\times m}^{\leq r}$ defines a birational equivalence, and since birational equivalence preserves dimension (due to the fact that it generates an isomorphism on the field of rational maps, which preserves the transcedence degree), then $\dim Z=\dim M_{n\times m}^{\leq r}$, showing $\dim M_{n\times m}^{\leq r}=r(n+m-r)$.

4 D

Problem 4

Hartshorne 1.1.10:

(a) If Y is any subset of a topological space X, then $\dim Y \leq \dim X$.

(b) If X is a topological space which is covered by a family of open subsets $\{U_i\}$, then $\dim X = \sup \dim U_i$.

(c) Give an example of a topological space X and a dense open subset U with $\dim U < \dim X$.

(d) If Y is a closed subset of an irreducible finite-dimensional topological space X, and if $\dim Y = \dim X$, then $Y = X$.

(e) Give an example of a Noetherian topological space of infinite dimension.

Solution:

(a) Let $n = \dim Y$, then there exists a maximal chain of irreducible closed subsets $Z_0 \subsetneq Z_1 \subsetneq ... \subsetneq Z_n \subseteq Y$ (under subspace topology). Now, for all index i, if consider $\overline{Z_i} \subseteq X$ (under topology of X), one has $Z_i = \overline{Z_i} \cap Y$ (since closure in the original space then take the intersection, is the same as taking the intersection then take the closure under subspace topology). So, $Z_i \subsetneq Z_{i+1} \implies \overline{Z_i} \subsetneq \overline{Z_{i+1}}$. Hence, we also have a chain of closed subsets $\overline{Z_0} \subsetneq \overline{Z_1} \subsetneq ... \subsetneq \overline{Z_n} \subseteq X$.

Finally, notice that this chain is consist of irreducible closed subsets, since Z_i is irreducible under subspace topology of Y implies it's irreducible under topology of X (since inclusion map $\iota : Y \hookrightarrow X$ is continuous, one must have Z_i irreducible getting sent to $\iota(Z_i) = Z_i$, which is also irreducible). So, using the fact that closure of an irreducible subset is irreducible, each $\overline{Z_i} \subseteq X$ is irreducible.

Hence, $\overline{Z_0} \subsetneq \overline{Z_1} \subsetneq ... \subsetneq \overline{Z_n} \subseteq X$ is a chain of irreducible closed subsets with length n in X, showing $\dim Y = n \leq \dim X$.

(b) Using part (a), it's clear first that for any $i \in I$ (the index set of $\{U_i\}$), one has $\dim U_i \leq \dim X$, hence one must have $\sup \dim U_i \leq \dim X$ (since $\dim X$ forms an upper bound of all $\dim U_i$).

Now, let $n := \dim X$, choose a maximal chain of irreducible closed subsets $Z_0 \subsetneq Z_1 \subsetneq ... \subsetneq Z_n \subseteq X$. Choose an open subset U_i such that $U_i \cap Z_0 \neq \emptyset$ (by the fact that $\bigcup U_\alpha = X$, such U_i exists). Then, since $Z_0 \subsetneq Z_j$ for any other index j, one also has $U_i \cap Z_j \neq \emptyset$.

Notice that $U_i \cap Z_j \subseteq Z_j$ is an open subset under subspace topology, then based on the assumption that Z_j is irreducible, any of its open subset must be irreducible and dense, so $U_i \cap Z_j$ is irreducible dense in Z_j for each index j, showing $\overline{U_i \cap Z_j} = Z_j$. (Note: since each Z_j is closed in X, such closure in X and Z_j coincides).

However, since $Z_j \subsetneq Z_{j+1}$, then with the fact that $\overline{U_i \cap Z_j} = Z_j$ and $\overline{U_i \cap Z_{j+1}} = Z_{j+1}$, one must have $U_i \cap Z_j \subsetneq U_i \cap Z_{j+1}$. Hence, in U_i (with subspace topology), one has the following chain of irreducible closed subsets:

$$U_i \cap Z_0 \subsetneq U_i \cap Z_1 \subsetneq ... \subsetneq U_i \cap Z_n \subseteq U_i \tag{4.1}$$

(Note: Each $U_i \cap Z_j$ is irreducible in Z_j, hence irreducible in X, which is also irreducible in any subspace containing it).

This shows that $\dim U_i = n = \dim X$, so $\dim X = \sup \dim U_i$.

(c) Consider the following topology on the set $\{1, 2, 3\}$:

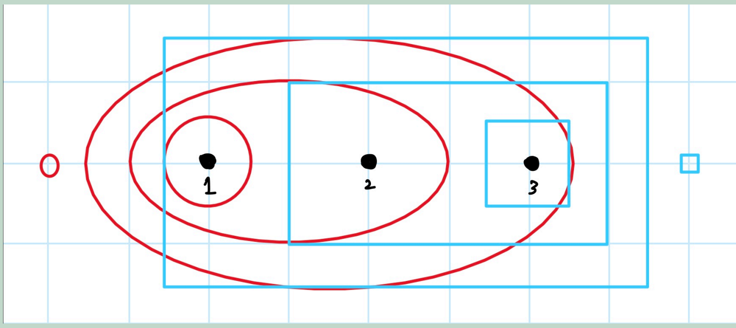

Abb. 1: Provided Topology on the 3-point set

Where, the red circles represent all the open sets, and the blue rectangles represent all the closed sets. Notice that every nonempty closed set C is irreducible, since the closed subsets for a proper chain of inclusion, so every proper closed subset of C unions to be a proper subset of C, so C cannot be written as unions of two proper closed subsets, hence irreducible.

First, notice that $X = \{1, 2, 3\}$ under this topology has $\dim X = 2$ (since counting nonempty irreducible closed subsets, there are total of 3 forming a proper chain of inclusions).

Now, consider the open subset $U = \{1, 2\}$ (the second largest red circle). It satisfies $\overline{U} = X = \{1, 2, 3\}$, since the only closed subset containing both 1 and 2 is $\{1, 2, 3\}$ (the largest blue rectangle). Also, if consider the subspace topology of U, the only closed sets are $\emptyset, \{2\}$, and $\{1, 2\} = U$ (again, both nonempty closed subsets are irreducible using the same claim as before). As a result, $\dim U = 1$ (since only two irreducible closed subsets exist, and form a proper chain of inclusion).

So, U is dense in X, while $\dim U = 1 < 2 = \dim X$.

(d) If Y is a closed subset of X, then any of its irreducible closed subset $Z \subseteq Y$ (under subspace topology of Y) is also an irreducible closed subset in X (since under inclusion map $\iota : Y \hookrightarrow X$, $\iota(Z) = Z$ is irreducible in X; while there exists closed set $C \subseteq X$, such that $Z = C \cap Y$, so Z is also closed under topology of X because Y is closed).

Then, suppose the contrary that $Y \neq X$, with X being irreducible, let $Z_0 \subsetneq Z_1 \subsetneq ... \subsetneq Z_n \subseteq Y$ be a maximal chain of irreducible subsets with length n (under subspace topology of Y), including into X, one has $Z_0 \subsetneq Z_1 \subsetneq ... \subsetneq Z_n \subsetneq X$ be a chain of irreducible subsets with length $n + 1$ (since X is irreducible, and $Y \subsetneq X$ implies $Z_n \subsetneq X$). So, it shows that $\dim X \geq n + 1 > n = \dim Y$, which contradicts the assumption that $\dim X = \dim Y$. Hence, $Y = X$ is enforced.

(e) Consider the natural number $\mathbb{N}$, and impose a topology so that the open sets are in the form $U_n := \{n + 1, n + 2, ...\}$ and empty set $\emptyset$ (which, including -1 so that $U_{-1} = \mathbb{N}$). Which, the closed sets are instead in the form $C_n := \mathbb{N} \setminus U_n = \{0, 1, ..., n - 1, n\}$.Pictorially, we have the following:

Abb. 2: Provided Topology on $\mathbb{N}$

Notice that for any ascending chain of open sets, say $U^{(1)} \subseteq U^{(2)} \subseteq ... \subseteq U^{(n)} \subseteq ...$, since each $U^{(n)} = U_{k_n}$ for some $k_n \in \mathbb{N}$, with $U_{k_n} \subseteq U_{k_{n+1}}$, it enforces $k_{n+1} \leq k_n$. So, since the defining integers satisfy $k_1 \geq k_2 \geq ... \geq k_n \geq ... \geq -1$, then it must stabilize at some $n \in \mathbb{N}$, showing for integer $l \geq n$, one has $k_l = k_n$, or $U^{(l)} = U_{k_l} = U_{k_n} = U^{(n)}$. Hence, all ascending chain of open subsets eventually stabilize, such topology makes $\mathbb{N}$ into a Noetherian space.

Also, every nonempty closed set C is irreducible, since the closed sets form a proper chain of inclusion, hence alll the proper closed subsets of C (where there's only finitely many) must union to be a proper closed subset, showing C can't be written as unions of proper closed subsets.

Finally, $\mathbb{N}$ under this topology has infinite dimension, simply because the irreducible closed subsets form an infinite strict ascending chain.